

Homework 2: Qualitative Analysis I — The Workflow

Paper track. Pairs with Home Lab 2. Do this on your own.

Part A: Identify from results (2 pts each)

For each unknown, give the **name and formula**.

#	Look	Water	pH	+ Acid	+ Iodine	Heat	Identity + formula
1	fine white powder	stays cloudy	~9	strong fizz	no change	just sits	
2	cubic crystals	dissolves fully	7	no fizz	no change	crackles, no melt	
3	fine white powder	"oobleck"	7	no fizz	blue-black	chars	
4	white powder	dissolves fully	~9	strong fizz	no change	just sits	
5	glassy grains	dissolves fully	7	no fizz	no change	melts, caramel smell	

Part B: Read the key (5 pts)

- | | | |
|-------------------------------------|-------------------|------------|
| 1. Fizzes strongly with acid? | yes → 2 | no → 3 |
| 2. Dissolves fully in water? | yes → BAKING SODA | no → CHALK |
| 3. Blue-black with iodine? | yes → CORNSTARCH | no → 4 |
| 4. Dissolves fully in water? | no → GYPSUM | yes → 5 |
| 5. Cubic crystals, no melt on heat? | yes → TABLE SALT | no → SUGAR |

Which numbered steps do you pass through, and what's the answer, for:

1. A powder that fizzes with acid and leaves cloudy undissolved residue.
2. A powder that doesn't fizz, is iodine-negative, dissolves fully, and melts to a brown syrup on heat.
3. A powder that doesn't fizz and turns iodine blue-black.

Part C: Build a key (4 pts)

You have four unknowns and know the list is: **Epsom salt, citric acid, corn starch, table salt**. Write a dichotomous key (first question, branches, down to all four). *Hint: which one test cleanly removes corn starch? Which removes citric acid?*

Part D: Mixture logic (2 pts)

A sample half-dissolves in water. The dissolved part is neutral and iodine- negative. The undissolved part turns iodine blue-black. Is this one powder or a mixture? Name the component(s).

Part E: Answer format (2 pts)

Rewrite this as a full scoring answer: "*Sample 4 is baking soda.*" (Assume the suspect who bakes is "Suspect B.")

Answer Key

Part A: 1. Calcium carbonate / chalk — CaCO_3 . 2. Table salt — NaCl . 3. Corn starch — $(\text{C}_6\text{H}_{10}\text{O}_5)_n$. 4. Baking soda — NaHCO_3 . 5. Sucrose (sugar) — $\text{C}_{12}\text{H}_{22}\text{O}_{11}$.

Part B: 1. Steps 1→2 → **chalk** (fizzes, doesn't dissolve). 2. Steps 1→3→4→5 → **sugar**. 3. Steps 1→3 → **cornstarch**.

Part C: One good key:

1. Turns iodine blue-black? yes → CORN STARCH no → 2
2. Water solution is acidic ($\text{pH} < 5$)? yes → CITRIC ACID no → 3
3. Crystals cubic (vs. needle-like), not bitter? yes → TABLE SALT no → EPSOM SALT

Part D: A mixture — table salt (soluble, neutral, iodine-negative) + corn starch (insoluble, iodine-positive).

Part E: "Sample 4: baking soda (sodium bicarbonate) — NaHCO_3 — implicates Suspect B, who bakes."